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Linear Algebra by Campus X.

Part : 1

1) What are vectors? Vectors are points on n-dimension space.

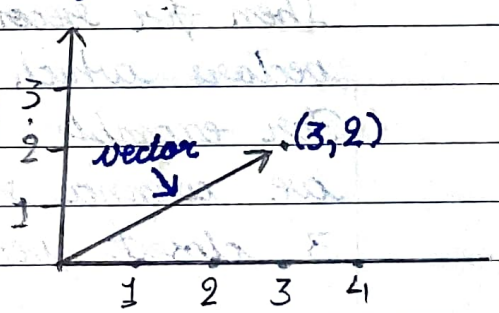
For example:

i) A point (4, 3) is lyinon on a 2-D space, then vector is [4, 3]; 4 : being x-component and 3 being the y-component.

ii) A point (3, 4, 5) lying on a 3-D space, then vector is [3, 4, 5] ; 3 being x component, 4 and 5 being y-component and z-component respectively.

n-dimension vector :

$[x_1, x_2, x_3, x_4 \dots x_n]$



2) Vectors from machine learning's view point :-

⇒ Consider iris dataset. See the sample below. We're having 150 datapoints.

SL	SW	PL	PW	specie
5.2	2.1	7.2	9.7	Versicolor
3.1	4.2	7	6.5	setosa
⋮	⋮	⋮	⋮	⋮

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each data point is fed into the model in the form of a vector.

Feture vectors
$v1 = [5.2, 2.1, 7.2, 9.7]$ → Model → Prediction → Setosa
$v2 = [3.1, 4.2, 7, 6.5]$ → Model → Prediction → Versicolor

We're having 150 flowers / datapoints. So we'll have 150 vectors in total representing each datapoint.

⇒ Example 2: Movie Recommendation system.

Supose we've 2 columns, movie and summary. Summary is the story of the movie in shat. We encode it first and then we plot the vectors into an n-dimension space.

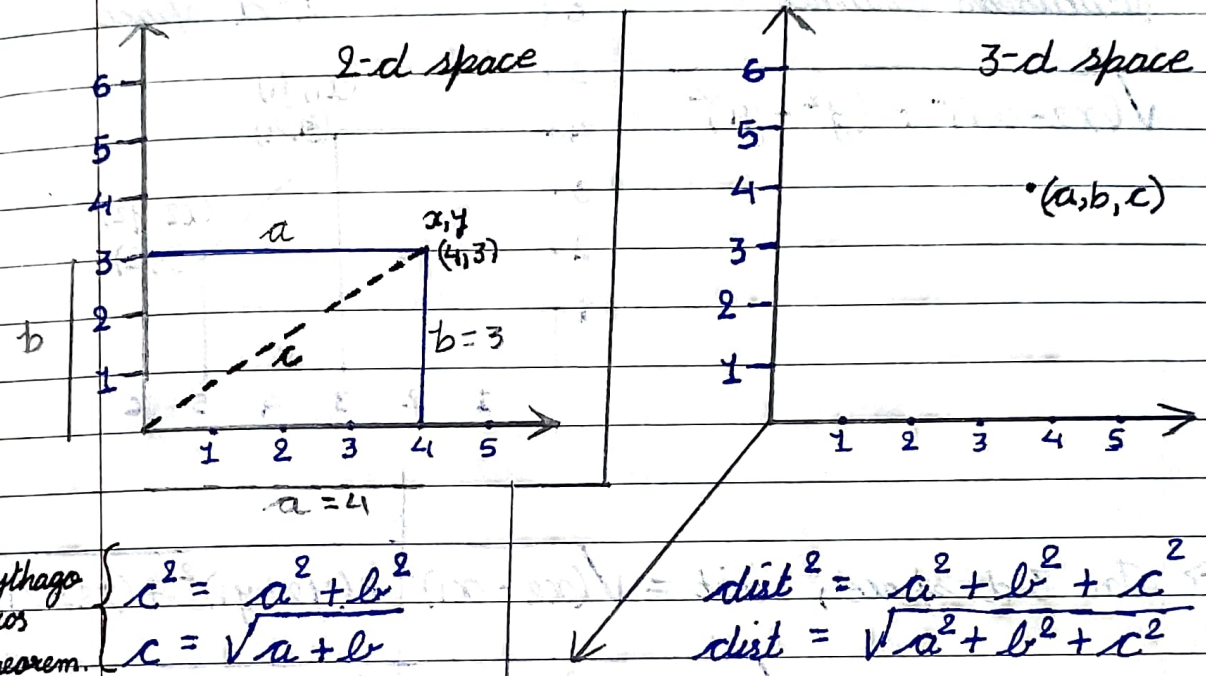
Then for recommendation, we recommend those vectors which are closer.
For example: Supose the movie is Yaarana and we wanna recommend 3 movies; we'll select the 3 closest vectors to Yaarana's vector.

3) Row and Column vector :-

- Row vector $x = [2, 4, 6, 8, 10]_{(1,5)}$
- Column vector $y = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \\ 9 \end{bmatrix}_{(5,1)}$

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4) Distance from origin :-



Pythagoras Theorem.

$$c^2 = a^2 + b^2$$
$$c = \sqrt{a + b}$$

$$dist^2 = a^2 + b^2 + c^2$$
$$dist = \sqrt{a^2 + b^2 + c^2}$$

$$\text{Distance in n-dimension} = \sqrt{x_1^2 + x_2^2 + x_3^2 + \dots + x_n}$$

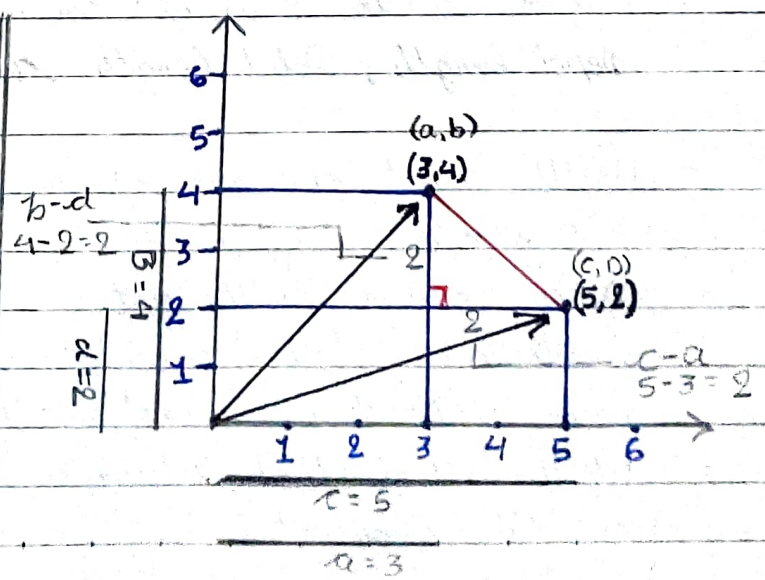
Note: Distance from the origin is also known as Magnitude of the vector.

5) Euclidean distance between two points:

$$\text{Euclidean distance} = \sqrt{(c-a)^2 + (b-d)^2}$$

$$= \sqrt{2^2 + 2^2}$$

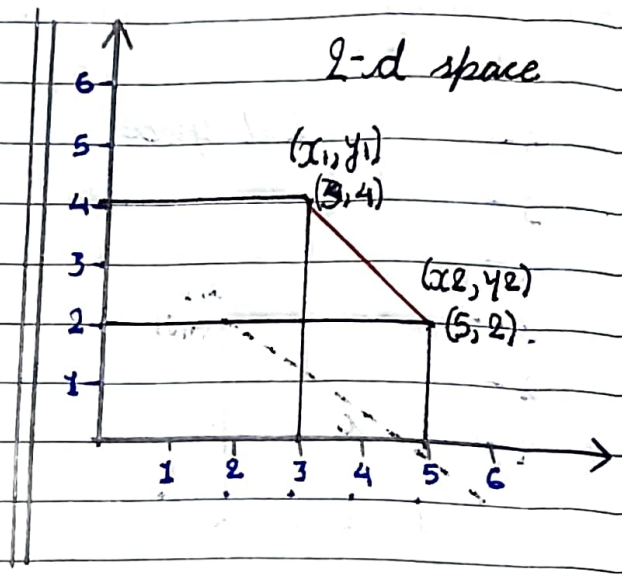
$$= \sqrt{8} = 2.83$$



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(2d-space)
Euclidean distance =

$$\sqrt{(x_2-x_1)^2+(y^2-y_1)^2}$$



→ In 3d-space, dist = $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$

→ In nD - dist = $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+\dots+(x_n-y_n)^2}$

⇒ : Euclidean distance in Machine Learning.

- In Machine Learning, there're various algorithms using Euclidean distance.
- One of the most famous is K-Nearest Neighbors.

Supose we've 2-D data (2 input columns and one target; Sepal length, Petal length and Specie.

- We'll plot vectors of each data point in a 2-D space.
- Now when we'll get a new data point for prediction, we'll calculate the distance of this new data point with all the datapoints.
- We'll select that vector's target whose distance is the shortest.

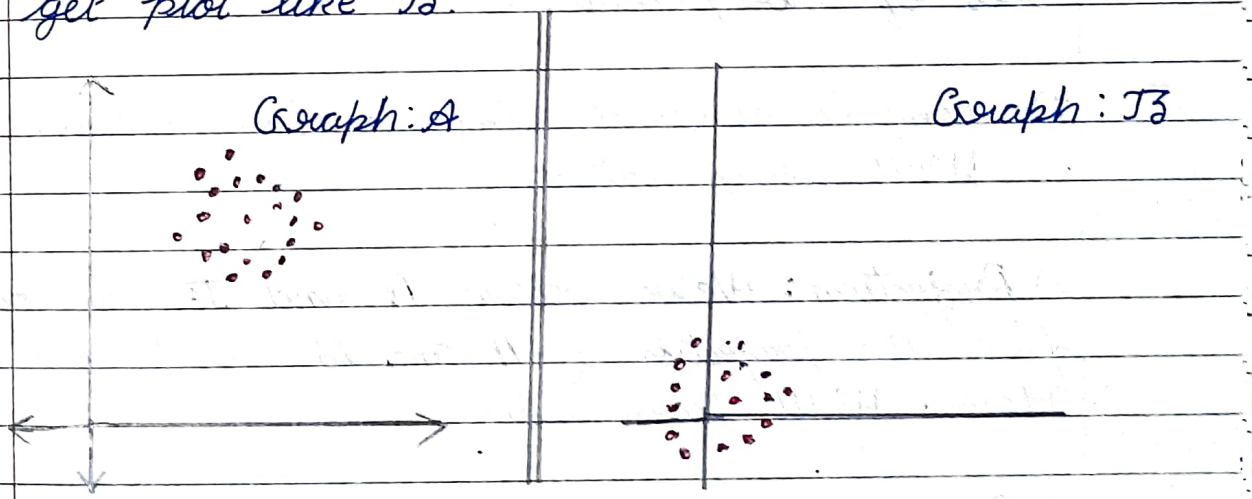
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6) b) scaler addition / substraction (shifting)

- $v1 = [2, 8, 4, 9] + 3$ (scalar addition) $= [5, 11, 7, 12]$
- $v2 = [3, 4, 9, 12, 14] - 2$ (scalar substraction) $= [1, 2, 7, 10, 12]$

⇒ Where is this used in Machine Learning ?

- It is used in mean centering.
- If we plot datapoints before and after deducting mean, before deduction we will get plot like A and after deduction we will get plot like B.



⇒ Some algorithms where mean centering is used. are: 1) PCA 2) Linear Regression 3) Gradient based optimizations 4) Clustering algo 5) Regurization

7) b) Dot product : When we multiply a vector with another vector, it is called a 'Dot Product' and the result is a scaler.

$A = [2, 4, 6, 8, 10]$ $B = [1, 3, 5, 7, 9]$

$A \cdot B = [2 + 12 + 30 + 56 + 90] = 190$

$A = [x1, x2, x3, \ldots, xn]$ $B = [y1, y2, y3, \ldots yn]$

$A \cdot B = [x1y1 + x2y2 + x3y3 + \ldots + xnyn]$

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Note: When we work with vectors and if not specified, by default we consider it as a column vector.

$$A = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \quad B = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$$

For $A \cdot B$, we do
i) $A^T B$ or $A \cdot B^T$

=> Rules of Dot product:-

1) Cumutative : $A \cdot B = B \cdot A$.
2) Distributive : $A(B+C) = AB + AC$

=> Uses of dot product :

1) Compare similarity between two vectors. How? We'll learn later.

2) Projection : We've vectors A and B, we can find the projection of A and B using dot product. How? We'll learn later.

3) Perform matrix multiplication.

=> Uses in Machine Learning :

1) Movie Recommender system :

- We plot vector for each data points in a graph.
- Then when we want 3 recommendation for movie Sholay, we find similarity score of Sholay's vector with each other vector and return top 3 with highest similarity score.

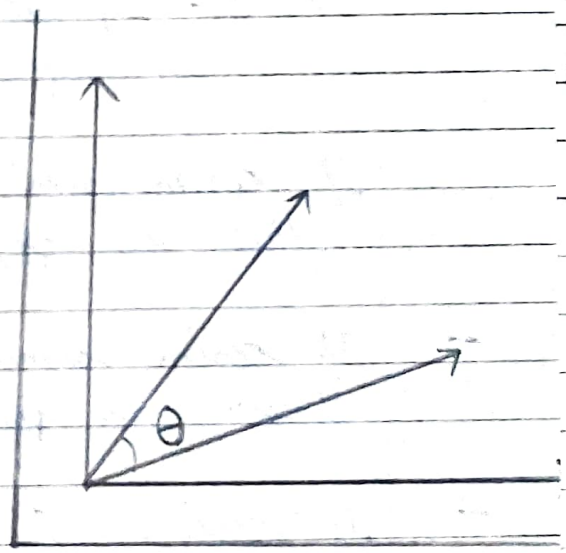
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- We use cosine similarity which we'll learn later

2) Deep Learning : In deep learning, we perform various matrix multiplications which simply means we require dot product.
- Means deep learning is totally relying on dot product.

8) Angle between two vectors.

- $A \cdot B = \|A\| \|B\| \cos\theta$



- Using dot product, we can conclude many features of vectors.
- For eg: If A and B are non-zero vector (means magnitude $\neq 0$) and it's given that $A \cdot B = 0$.
In this case, we can conclude that $\|A\| \|B\| \cos\theta = 0$ and $\theta = 90°$ and $A \perp B$.
This suggests that A and B are dis-similar vector

- Using dot product, we can calculate angle between two vectors.
Supose components of $A = [a_1, a_2, a_3 \dots, a_n]$ and $B = [b_1, b_2, b_3, \dots b_n]$

$$\therefore \cos\theta = \frac{A \cdot B}{\|A\| \|B\|}$$

$$\therefore \theta = \cos^{-1}\left[\frac{A \cdot B}{\|A\| \|B\|}\right]$$

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9) Cosine similarity : Used to measure similarity of two vectors.

- $\cos\theta = \frac{AB}{\|A\| \|B\|}$; means cos of angle between A and B.
- Cosine similarity gives result between -1 and 1
- If $\theta = 0$, it means the vectors are of same direction and are totally similar.
- If cosine similarity = (-1), means vectors are totally in opposite direction and hence $\theta = 180^\circ$.
- If cosine similarity = 0, means $\theta = 90^\circ$ and hence there's no similarity.
- If cosine similarity is between 0-1, θ = acute angle and similarity increases from 0-1
- If cosine similarity is between -1-0, θ = obtuse angle.
- We can use cosine similarity for Movie Recommendation system.

Note: In last pages, we've learnt

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Now we'll learn the eqⁿ of hyperplane but before that, let's learn few things.

Calculate regression line for below given data.

x	y	x^2	y^2	xy
0	5	0	25	0
1	3	1	9	3
2	2	4	4	4
3	0	9	0	0
4	-1	16	1	-4
5	-3	25	9	-15
6	-5	36	25	-30
7	-6	49	36	-42
8	-8	64	64	-64
9	-9	81	81	-81
45	-22	285	254	-229

$\bar{x} = \frac{45}{10} = 4.5$

$\bar{y} = \frac{-22}{10} = (-2.2)$

$\Rightarrow y = mx + c$; where

i) $m = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$

ii) $c = \bar{y} - m\bar{x}$

$\Rightarrow$ Slope $m = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$

$= \frac{(10)(-229) - (45)(-22)}{(10)(285) - (45)^2}$

$= \frac{(-2290) + 990}{2850 - 2025} = \frac{-1300}{825} = \boxed{(-1.5758)}$

$\Rightarrow$ Intercept $c = \bar{y} - m\bar{x} = (-2.2) - (-1.5758)(4.5)$

$= (-2.2) + 7.0911$

$= 4.89$

$= \boxed{4.8911}$

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∴ Regression line :

$$y = mx + c$$
$$= (-1.5758)x + 4.8911$$

⇒ Convert regression eq^n in $Ax + By + C = 0$ form.

$$y = -1.5758x + 4.8911$$

$$\boxed{-1.5758x - y + 4.8911 = 0}$$

'OR'

$$\boxed{1.5758x + y - 4.8911 = 0}$$

i) Slope $= \frac{-A}{B} = \frac{-1.5758}{1} = (-1.5758)$

ii) Intercept $= \frac{-C}{B} = \frac{-(-4.8911)}{1} = 4.8911$

Note : Both above equations are same. We'll get same slope and intercept of both the equations.
- If we multiply eq^n 1 with (-1) we'll get eq^n 2

★ What is slope and intercept ?

1) Slope :

→ Consider this data. $x = [0, 1, 2, 3, 4, 5]$
$y = [4, 2, 0, (-2), -4, -6]$

→ Slope determines 'rate of change'
→ Here, in the above data slope is (-2).
→ Which simply means if x increases by 1, y decreases by 2. $x \uparrow 1 \rightarrow y \downarrow 2$.

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→ In regression line, the effect of slope is that it tilts the line.

Visual meaning :
- slope = 2 → Steep uphill
- slope = (-2) → Steep downhill
- slope = 0 → Flat road.

We got slope (-1.5758), means $x \uparrow 1$, $y \downarrow (-1.5758)$

Why is the slope useful ? It tells us trend / relationship between x and y.

→ Remember, slope = rate of change in y, when x changes.

2) Intercept :

→ Intercept determines the point where the regression line cuts y-axis.
→ This simply means, what is y-coordinate when $x = 0$.
→ That is the point where line of cuts y-axis.
→ That point is the starting point.
→ In our previous data, intercept was 4.8911. Means at $x = 0$, $y = 4.8911$.
→ So intercept = 4.8911 and starting point = (0, 4.8911)

Real life meaning of intercept.
→ It tells us the starting value.
For example :
1) Salary growth → Intercept → Starting salary.
2) Distance → Intercept → Initial position.
3) Bank balance → Intercept → Initial balance.

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Final value slope intercept
$y = mx + c$

$\therefore$ Final value = (rate × input) + initial value
(rate × input) = slope; initial value = intercept

We know that the slope shows rate of change but why do we need intercept?

→ Because a slope alone doesn't define a line.
→ Slope tells us the rate of change but what about the starting point?
→ For example:
i) $y = 2x + 4$
ii) $y = -2x + 10$
iii) $y = -2x - 3$
All eq^n have same slope (-2) but different line positions if we plot.

So what does an intercept do?
→ Intercept fixes the position of the line.
→ Slope tilts the line whereas intercept moves line up and down.
→ Specifically it tells us where the line cuts y-axis when $x = 0$.

Real life intution:
$y = mx + c$

V. 1) Think of salary growth, m shows yearly increament and c shows the starting salary.
Without c we know the growth rate but not where we started.
2) Distance: m = speed and c = starting position
Without c we won't know from where did we start.

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⇒ Overview:

- Slope = rate of change in y when x changes by 1 unit.
- It tilts the regression line.
- Intercept = starting point. Shows from where we started.
- It moves the line upwards and downward.

☆ Equation of a hyperplane.

Now our main topic starts. Consider eq^n.

$y = mx + c$.

General eq^n: $Ax + By + C = 0$

$By = -Ax - C$

$\therefore y = \boxed{\frac{-A}{B}} x \boxed{\frac{-C}{B}}$; $m = \frac{-A}{B}$, $c = \frac{-C}{B}$

Slope(m) Intercept(c)

$Ax + By + C = 0$

$(x, y) = (2, 3) \rightarrow A2 + B3 + C = 0$

$(x_1, x_2) = (2, 3) \rightarrow Ax_1 + Bx_2 + C = 0$

Let $A = W_1$, $B = W_2$ and $C = W_0$, so eq^n become

$\therefore W_1x_1 + W_2x_2 + W_0 = 0$

$\rightarrow$ Slope $= \frac{(-W_1)}{W_2}$ and Intercept $= \frac{(-W_0)}{W_2}$.

2-D: $W_1x_1 + W_2x_2 + W_0 = 0$

3-D: $W_1x_1 + W_2x_2 + W_3x_3 + W_0 = 0$

n-D: $W_1x_1 + W_2x_2 + \cdots + W_nx_n + W_0 = 0$

this is the end.

and all W's are packed in W^T and hence W^T also hits the line.

- W_0 moves line upward and downward which intercept also does. This is true. Date: Page No:

intercept $= \frac{-W_0}{W_2}$ and since W_0 is the part of the eq^n it moves changes the line upward and downward.

$\Rightarrow$ Convert into vector forms.

- Vector $W = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix}_{n\times 1}$, vector $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}$

- Here dot product is not possible because shape of $W = n\times 1$ and $x = n\times 1$, where no. of cols of W and no. of rows of x are not same.
- Therefore we transpose W.

$W^T = \underset{1\times n}{[w_1, w_2, w_3 \cdots w_n]}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}$

$\therefore$ eq^n $w_1x_1 + w_2x_2 + \ldots + w_nx_n + w_0 = 0$ becomes

$$\boxed{W^Tx + W_0 = 0}$$ Applicable to N-D data.

Tip: In $w_1x_1 + w_2x_2 + \ldots + w_nx_n + w_0 = 0$, W represents weights for each feature i.e x_1, x_2, x_n. x represents features of the data and w_0 is bias term. (constant).

- w_0 is also an intercept.

$Ax + By + C = 0$ converted to $w_1x_1 + w_2x_2 + w_0 = 0$

- $w_1 = A$ = Weight of feature x or x_1
- $w_2 = B$ = Weight of feature y or x_2.
- $w_0 = C$ = Intercept / Bias (constant term) because we rearranged $y = mx + c$ to $Ax + By + c$.
- Originally, slope $= \frac{-A}{B} = \frac{-w_1}{w_2}$ and intercept $= \frac{-C}{B} = \frac{-w_0}{w_2}$
- As we rearranged the eq^n $y = mx + c$, we get weight of $y = (-1)$ or (y) and hence C = intercept. But regression line's eq^n was always $y = mx + c$.